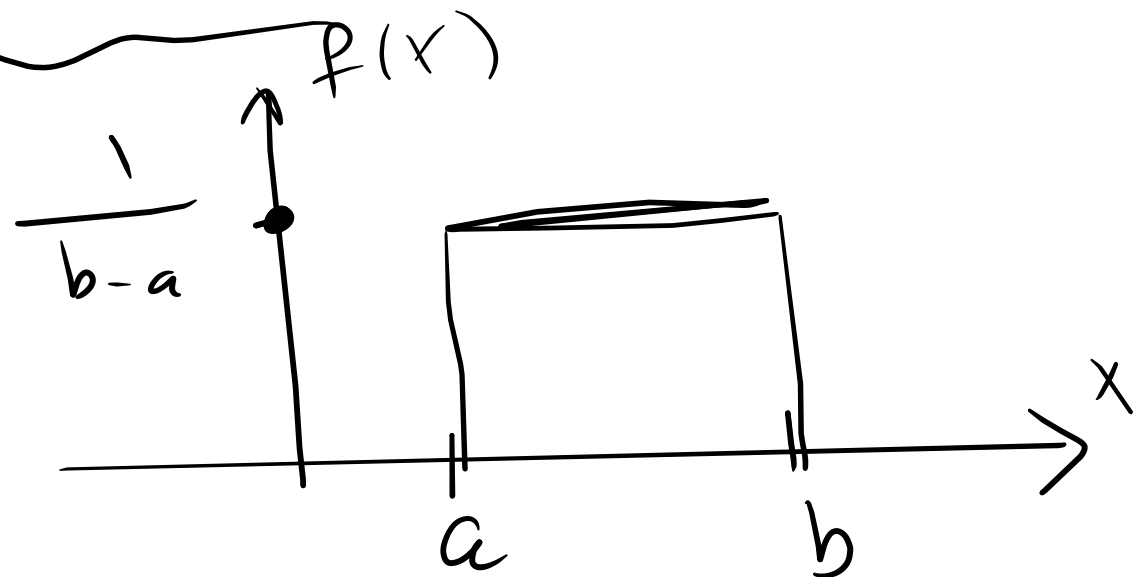


Uniform distribution

$$f(x) = \begin{cases} 0 & x < a \\ \frac{1}{b-a} & a \leq x \leq b \\ 0 & x > b \end{cases}$$



$$\text{mean} = \mu = \int_a^b x f(x) dx = \int_a^b x \frac{1}{(b-a)} dx$$

$$= \frac{1}{(b-a)} \left. \frac{x^2}{2} \right|_a^b = \frac{1}{2(b-a)} [b^2 - a^2]$$
$$= \frac{1}{2(b-a)} (b-a)(b+a)$$

$$\boxed{\mu = \frac{a+b}{2}}$$

$$\sigma^2(x) = \int_a^b x^2 f(x) dx - \mu^2$$

Uniform
Dist

$$= \int_a^b x^2 \left(\frac{1}{b-a} \right) dx - \mu^2$$

$$= \frac{1}{(b-a)} \left[\frac{x^3}{3} \right]_a^b = \frac{1}{3(b-a)} [b^3 - a^3]$$

$$= \frac{1}{3(b-a)} (b^2 + ab + a^2) = \frac{a^2 + ab + b^2}{3}$$

$$= \frac{a^2 + 2ab + b^2 - ab}{3} = \frac{1}{3} [(a+b)^2 - ab]$$

$$\sigma = \sqrt{\frac{1}{3} (a+b)^2 - ab}$$

A continuous rv X is said to have a **uniform distribution** on the interval $[A, B]$ if the pdf of X is

$$f(x; A, B) = \begin{cases} \frac{1}{B - A} & A \leq x \leq B \\ 0 & \text{otherwise} \end{cases}$$

Example 4.5

"Time headway" in traffic flow is the elapsed time between the time that one car finishes passing a fixed point and the instant that the next car begins to pass that point. Let X = the time headway for two randomly chosen consecutive cars on a freeway during a period of heavy flow. The following pdf of X is essentially the one suggested in "The Statistical Properties of Freeway Traffic" (*Transp. Res.*, vol. 11: 221-228):

$$f(x) = \begin{cases} .15e^{-.15(x-.5)} & x \geq .5 \\ 0 & \text{otherwise} \end{cases}$$

④ $P(X \leq 5)$

$$f(x) = \begin{cases} 0.15 e^{-0.15(x-.5)} & , x \geq 0.5 \\ 0 & \end{cases}$$

① prove that $f(x)$ is p.d.f ② $\mu = ?$
 $\sigma^2 = ??$ ③ $P(X \geq 2)$

$$f(x) = \begin{cases} .15e^{-.15(x-.5)} & x \geq .5 \\ 0 & \text{otherwise} \end{cases}$$

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_{.5}^{\infty} .15 e^{-.15(x-.5)} dx$$

$$= .15 \int_{.5}^{\infty} e^{-.15(x-.5)} dx$$

$$= -0.15 e^{-.15(x-.5)} \Big|_{.5}^{\infty}$$

$$= -[e^{-\infty} - 1] = -[0 - 1] = 1$$

~~##~~ p.d.f

$$\begin{aligned} \text{mean} = E(X) &= \int_{0.5}^{\infty} x f(x) dx = \int_{0.5}^{\infty} x (\underline{0.15}) e^{-0.15(x-0.5)} dx \\ &= \left[-\textcircled{x} e^{-0.15(x-0.5)} - \frac{1}{0.15} e^{-0.15(x-0.5)} \right]_{0.5}^{\infty} \end{aligned}$$

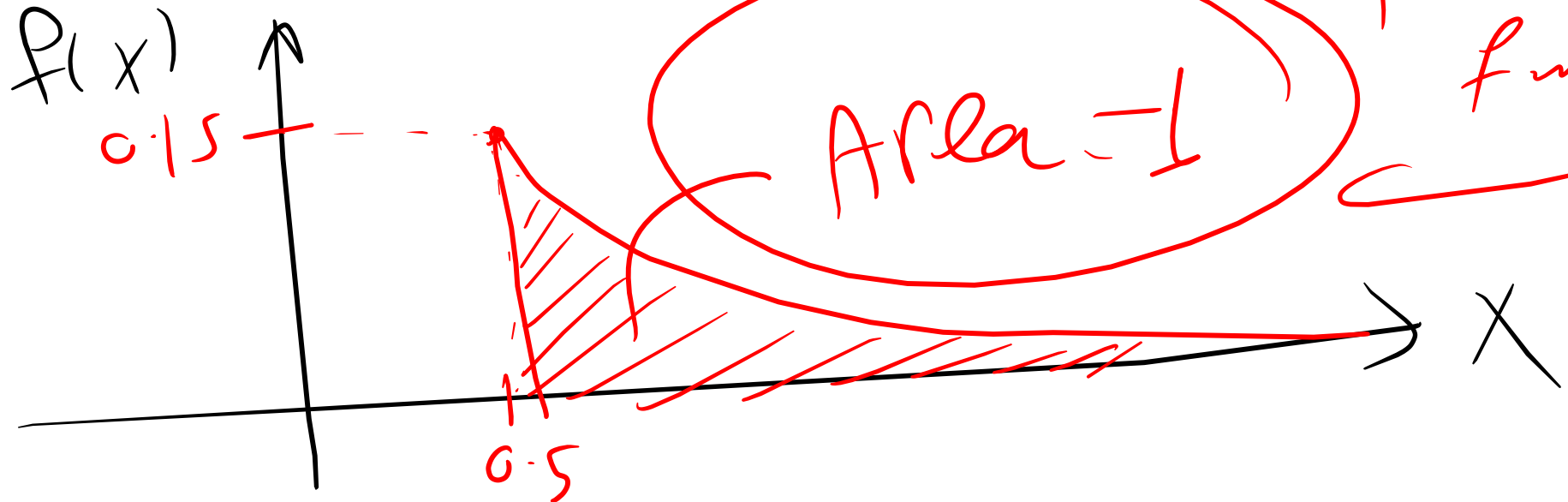
$$= \infty \cdot 0 - \frac{1}{0.15} (0 - 1)$$

$$\mu = 6.65 = \underline{\underline{\frac{20}{3}}}$$

$$\begin{aligned} &0.15 e^{-0.15(x-0.5)} \\ &- e^{-0.15(x-0.5)} \\ &= \frac{1}{0.15} e^{-0.15(x-0.5)} \end{aligned}$$

$$\sigma^2 = \int_{0.5}^{\infty} x^2 \left(0.15 e^{-0.15(x-0.5)} \right) dx - \left(\frac{20}{3} \right)^2$$

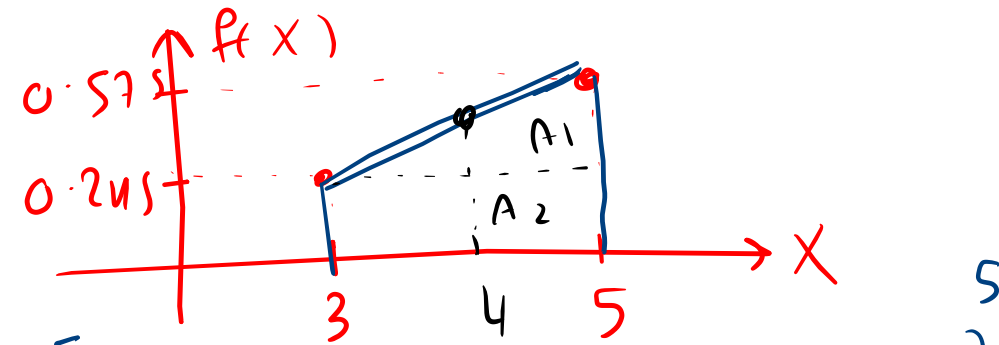
$$S.D = \sigma = \sqrt{\sigma^2}$$



1. The current in a certain circuit as measured by an ammeter is a continuous random variable X with the following density function:

$$f(x) = \begin{cases} .075x + .2 & 3 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

- Graph the pdf and verify that the total area under the density curve is indeed 1.
- Calculate $P(X \leq 4)$. How does this probability compare to $P(X < 4)$?
- Calculate $P(3.5 \leq X \leq 4.5)$ and also $P(4.5 < X)$.



$$\textcircled{a} \int_3^5 (.075x + .2) dx = \left[\frac{.075}{2} x^2 + .2x \right]_3^5 = 1$$

$$\textcircled{b} \text{ mean} = \int_3^5 x(.075x + .2) dx = \left[\frac{.075}{3} x^3 + \frac{.2}{2} x^2 \right]_3^5 = \checkmark$$

$$P(X \leq 4) = \int_3^4 (.075x + .2) dx$$

$$\textcircled{c} P(3.5 < X < 4.5) = \int_{3.5}^{4.5} (.075x + .2) dx = \frac{1}{2}$$

4. Let X denote the vibratory stress (psi) on a wind turbine blade at a particular wind speed in a wind tunnel. The article “Blade Fatigue Life Assessment with Application to VAWTS” (*J. of Solar Energy Engr.*, 1982: 107–111) proposes the Rayleigh distribution, with pdf

$$f(x; \theta) = \begin{cases} \frac{x}{\theta^2} \cdot e^{-x^2/(2\theta^2)} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

as a model for the X distribution.

- Verify that $f(x; \theta)$ is a legitimate pdf.
- Suppose $\theta = 100$ (a value suggested by a graph in the article). What is the probability that X is at most 200? Less than 200? At least 200?
- What is the probability that X is between 100 and 200 (again assuming $\theta = 100$)?
- Give an expression for $P(X \leq x)$.